Health Chatbot Using Natural Language Processing for Disease Prediction and Treatment

Philip Indra Prayitno
Computer Science Department, School of Computer Science
Bina Nusantara University
North Jakarta, Indonesia
philip.prayitno@binus.ac.id

Reinhart Perbowo Pujo Leksono
Computer Science Department, School of Computer Science
Bina Nusantara University
Bogor, Indonesia
reinhart.leksono@binus.ac.id

Fernando Chai
Computer Science Department, School of Computer Science
Bina Nusantara University
Pontianak, Indonesia
fernando.chai@binus.ac.id

Richard Aldy
Computer Science Department, School of Computer Science
Bina Nusantara University
West Jakarta, Indonesia
richard.aldy@binus.ac.id

Widodo Budiharto
Computer Science Department, School of Computer Science
Bina Nusantara University
West Jakarta, Indonesia
wbudiharto@binus.edu

2021 1st International Conference on Computer Science and Artificial Intelligence (ICCSAI) | 978-1-6654-4002-8/21/$31.00 ©2021 IEEE | DOI: 10.1109/ICCSAI53272.2021.9609784

***Abstract*—People who don't know about products or services provided by a company need a system that can provide answers to the questions that are usually asked. This system is called Frequently Asked Questions. This system is not effective and efficient in providing the information. In this paper, we propose a chatbot using Natural Language Processing to provide some information about health.**

Here, we use NLP to make a chatbot system that can understand and answer the questions that are asked by the user. The cosine similarity is used to find the similarities between the query words (the questions that are asked by the user) and the documents, then return the answers of the document with the highest similarity. Based on our development, our medical chatbot has successfully diagnose the user illness with approximately 87 percent accuracy.

Keywords—Artificial Intelligence, Chatbot, Cosine Similarity, ID3 Decision Tree, Natural Language Processing.

I. INTRODUCTION

A chatbot is a computer program that conducts a conversation via auditory or textual methods. These programs are designed to provide a clone of how a human will chat and thereby it acts as a conversational partner rather than humans [1]. They provide a simulating platform for effective and smart communications with the user on the other end.

The chatbot was developed in the 1960s. Initially, chatbots were an experimental computer program to trick people into chatting with humans when in fact they were talking to a machine. As time goes by, the chatbot continues to experience progress. Nowadays, the chatbot has begun to spread into many different fields of study, including health.

Natural Language Processing (NLP) is a widely discussed and researched topic nowadays. As it is one of the oldest areas of research in machine learning it is used in major fields such as machine translation, speech recognition and text processing. In our chatbot, we will implement the concept of cosine similarity from natural language processing so that it can return the best solution for the patient.

Naturally, when someone is in an unhealthy condition they go to the hospital or health clinics to see the doctors. The common problem we usually face is "what if we didn't have time to go to the hospital?" followed by other problems. There must be a solution to those problems.

In these modern days, there are things called online consultation or online health checkups with a doctor which is considered as a solution to all of those problems. But once again there's still a problem with that solution, we have to make a reservation to make sure that the doctor is available, we absolutely couldn't wait that long.

Artificial intelligence provides the greatest ability to imitate human thought and behaviour in front of a machine [2]. If we use all of the knowledge from doctors and convert it to a database. We can take that advantage to create a chatbot that is working just like a doctor in general and that will also solve all the existing problems.

II. LITERATURE REVIEW

A. Chatbot

Darius and Sophie [3] have noted that the word "Chatbot" is formed from the terms "chat" and "robot". At first, the term chatbot is used for a computer program that simulates human language with the help of text-based dialog systems. Chatbot contains input and output texts that allow users to communicate with the chatbot software, such that it gives a feeling of talking to a real person.

As Gajendra and company have said that [4] a chatbot has to be trained first so that it can remember different kinds of data structure like an array, queue, stack, tree, etc. for it to be able to respond to the user according to its knowledge.

The statement above indirectly supports Divya and the company's claim that [5] at the basic level there are 2 types of a chatbot, that is unintelligent (using predefined conversation) and intelligent (using machine learning). A chatbot can be made using different algorithms. However, the frequently used algorithm in general is Support Vector Machine (SVM) Classifier, Naïve Bayes, and K-Nearest Neighbor.

B. Artificial Intelligence

Haristiani stated that [6] Artificial Intelligence (AI) is the foundation of every kind of chatbot that is combined with NLP so that a chatbot responds according to data obtained from the user. Nadarzynski and colleagues also stated [7] that AI is a software that is comprised of complex mathematical algorithms that process information to generate an output that has been predetermined, which leads to a relevant result.

978-1-6654-4002-8/21/$31.00 ©2021 IEEE

Licensed use limited to: Alvas Institute of Engineering and Technology Department of Library and Information Centre. Downloaded on March 30,2026 at 13:31:08 UTC from IEEE Xplore. Restric

Martin et al. [8] have stated that chat service agents have been mostly replaced by conversational software agents or chatbot and they are designed to communicate with users using natural language that is based on AI; even though it saves budget and time, implementation of chatbot AI often fails to meet the expectation of customers, potentially resulting in users to not follow the requests from the chatbot.

C. Natural Language Processing

As we know, computer systems cannot directly understand human language. That's why Mathew and colleagues stated that [9] Natural Language Processing (NLP) can help computer systems to understand the language that humans use and classify them, as well as analyze them if the dialog needs a response.

One of the important functions of NLP is Tokenization, removing stopwords, and lemmatization. Lekha and colleagues describe tokenization as [10] the process of breaking down a sentence into words and removing punctuations that in the end removing stop words will be done, that is removing unimportant words so we can gather keywords for the next processes. Chaitrali et al. have mentionted that [11] removing stopwords is used to remove words that are often used like 'is', 'like' etc. Lemmatization is then used to obtain the base word of each token.

D. Cosine Similarity

As stated before, keywords that are obtained from removing stop words will be measured their similarity with pre-existing documents using the Cosine Similarity formula as shown below.

$$|D_i| = \sqrt{\sum_i w_{i,j}^2} \qquad |Q| = \sqrt{\sum_i w_{Q,j}^2}$$

Where:
$|D_i|$ = Vector Length of Document i^{th}
$W_{i,j}$ = Weight of Word j^{th} in Document i^{th}
$|Q|$ = Vector Length of Query

The vector length of each document will then be used to get the dot product value. Then by using the dot product value, vector length of document i^{th} and vector length of query, we could get the best result from the query by looking at the highest value of the calculation.

Bo-Hao and colleagues have combined the use of Cosine Similarity with Recurrent Neural Network (RNN) algorithm and TF-IDF [12] to predict the user's illness and what medical product will be perfect for them. From their experiment, they managed to acquire 88% accuracy.

Aside from that, Cosine Similarity for chatbot development is also used by Mayuresh and colleagues [13] in an attempt to develop a more natural usage of sentences by a chatbot using word embedding technique. In short, Cosine Similarity is an algorithm to measure the likeness of 2 objects.

E. ID3 Decision Tree

Iterative Dichotomiser 3 (ID3) Decision tree is an algorithm to generate a decision tree. ID3 Decision tree is a simple technique to make choices. They are used in many things, from animation to strategic AI.

The ID3 algorithm was proposed by Quinlan Ross in 1986. The tree is made from top-down while selecting attributes in each node. In the algorithm, an Information Gain value is used to determine which attribute to use. According to Shamrat et al, [14] The algorithm will stop when Information Gain reaches 0 or every case has been used. Based from the Shamrat et al experiment [14], ID3 Decision tree has an accuracy around 52% for prediction result.

ID3 Decision tree is used in a chatbot to enhance their performance. Mujeeb and colleagues also stated that [15] from a certain information, a decision tree takes action that it thinks is the most appropriate from a collection of actions. The problem in making a decision tree is finding what action to choose. The handling of this problem is called attribute selection. There are a lot of ways to do attribute selection. According to Panhalkar and Doye, Some popular methods are [16] ID3, CART, and C4.5

III. Proposed Method

A. Workflow

The bot will inherit common doctors' consultation behavior. First of all, the chatbot will ask whether the user already knows their disease. If they do, the user could simply input the disease they suffer from. Then, it will look for the solution for said disease in the database.

If the user doesn't know about what kind of disease that they have, the user could describe all of his symptoms one by one, then the bot will perform stop word removal and compare the user's input with the available symptoms in the database using cosine similarity. The symptom with the highest value is then used for the ID3 decision tree algorithm, to find the disease as quickly as possible. If the bot manages to find the disease, it will give the disease treatment to the user. And of course, it will output a disclaimer.

B. Algorithm

We all know that every chatbot has at least 3 main steps inside its algorithm. First, user will input any kind of text to the chatbot. Second, the chatbot will then starts to analyze and process the user input. And lastly, it will produce an output that contains the response according to the user input.

In our chatbot, we've done some modification to the core algorithm as shown in the algorithm below.

```
while true
    declare userDisease
    if userDidntKnowTheirDisease then
        while diseaseIsNotFound then
            getUserSymptom
            doUserInputStopWordRemoval
            if diseaseIsNotFound and endOfFile
                break
            end if
```

Licensed use limited to: Alvas Institute of Engineering and Technology Department of Library and Information Centre. Downloaded on March 30,2026 at 23:31:08 UTC from IEEE Xplore. Restric

```
            end while
        else
            doUserInputStopWordRemoval
            userDisease = getKeywordsFromUserInput
        end if
        getDiseaseSolutionFromDatabase
        showUserDiseaseAndSolution
        showDisclaimer
    end while
```

From the pseudocode above, we can see that the program will keep looping until the user close it. First the program will ask whether the user already know their disease or not.

If they do, then user can simply input their disease and the program will do stop word removal and try to find the query on the database. Then it will simply show their disease with the treatment including a disclaimer message.

If they don't, then user can input their symptoms one by one, and the program will look towards the matched disease based from the user symptoms. And finally, it will show their disease, with the treatment and also the disclaimer message.

C. Flowchart

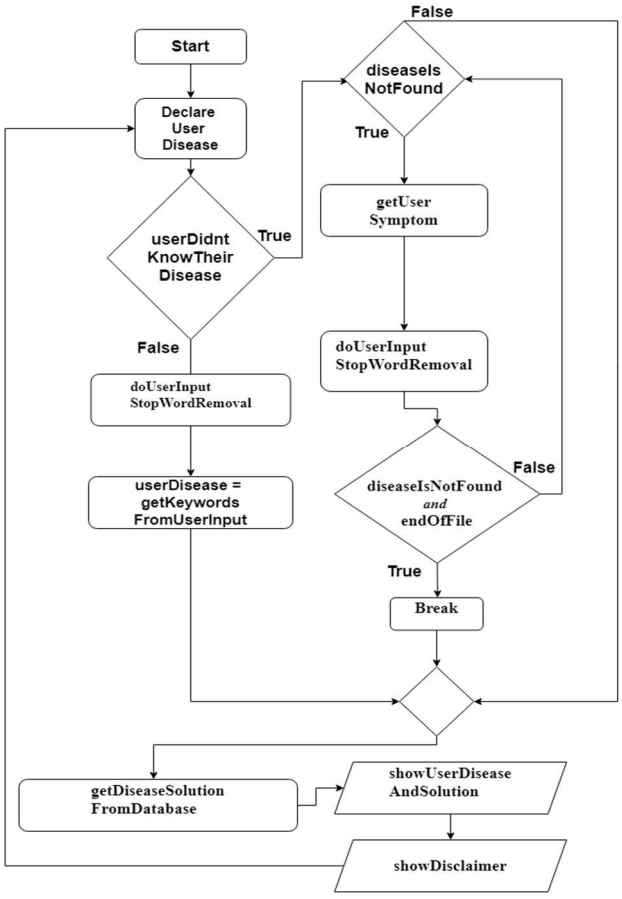


Fig. 1. Application Flowchart

As shown in Figure 1. We can see the visual representation of the algorithm in point B where it's constructed using flowchart.

D. Application Prototype

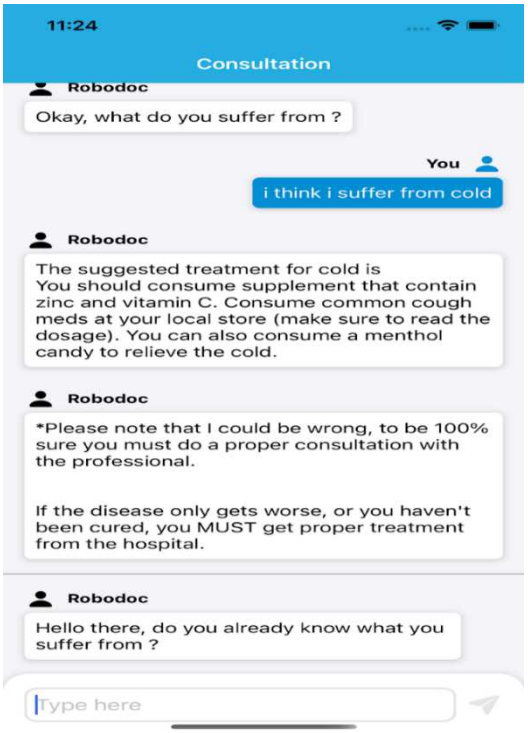


Fig. 2. Application Prototype

User can interact with the chatbot by inputting their disease or symptoms inside the textbox. As shown in Figure 2. The chatbot has successfully analyze the user input and provide the treatment referring to the user disease.

IV. Experimental Results

This experiment is using data from the user inputs. The experiment is going to be divided into the following steps: first, the bot asks the user whether they already know the disease they suffer from. Second, if the user does not know the disease, the bot will ask the user about symptoms they have, but if the user does already know the disease, they can simply type it. Last one, the bot will show what user suffer from and get the treatment for user disease.

A. Results

First, we collect a training data set that contains all of the disease information including the symptoms and also its treatment from the alodokter official website (a medical website that has been assured and approved by the Indonesian Health Minister). For the testing purposes we managed to collect 113 disease data with its treatment and 640 symptoms data according to the disease.

After the training data is collected, we proceed to begin in developing a chatbot based on the iOS mobile application. After the application development process is finished, we

Licensed use limited to: Alvas Institute of Engineering and Technology Department of Library and Information Centre. Downloaded on March 30,2025 at 23:31:08 UTC from IEEE Xplore. Restric

conduct a several test with these following user input and we managed to receive the following output as shown in Table 1. and Table 2.

TABLE 1. THE DATA RESULT (USER ALREADY KNOW THEIR DISEASE)

No	User Input	Output
1	i think i suffer from cold	The suggested treatment for cold is You should consume supplements that contain zinc and vitamin C. Consume common cough meds at your local store (make sure to read the dosage). You can also consume menthol candy to relieve the cold.
2	what should i do if i have covid	The suggested treatment for covid is Isolate yourself in about 2 weeks, drink water, get enough rest, and consume pain reliever to relieve the cough, fever and pain. If you didn't get any better make sure to do further check up at the hospital.
3	i think i'm crazy	Sorry, I have tried my best and didn't come up with a solution to that disease(crazy)**.**
4	help, my uncle suffer from epilepsy	The suggested treatment for epilepsy is Consume anti epilepsy medicine such as carbamazepine, clobazam, levetiracetam, etc., to stabilize your electrical neuron activity in your brain.
5	what should i do if my friend get asthma	The suggested treatment for asthma is that you should always bring a relief inhaler. If the asthma is too chronic, get medical treatment at the hospital as soon as possible.
6	i got a very big boil	The suggested treatment for bigboil is Compress the boil with warm water 3 times a day. And when the boil is broken, make sure to clean them with sterile gauze and antibacterial soap, then cover the boil with sterile gauze.
7	my hands got burned	The suggested treatment for burned hands is To treat burns you have to extinguish the fire or remove the burning object from your body, followed by removing the clothes or cloth that is attached to your burned skin. Next, you can cool the burn with running water and take paracetamol to relieve the pain. If the burn is severe, you are advised to receive further treatment at the hospital.
8	my father keep coughing all the time	The suggested treatment for coughing is You should rest and. - Try to take a sip of honey - Consume common cough meds at your local store (make sure to read the dosage) - If the cough still exists, it may be because of bacteria (you should visit a clinic or hospital) to get antibiotics.

TABLE 2. THE DATA RESULT (USER DIDN'T KNOW THEIR DISEASE)

No	User Input	Predicted User Disease (Output)	Actual Disease
1	i have a red spot all over my skin, i keep sneezing all the time	According to my analysis, you have been potentially suffered from Allergy.	Allergy
2	sometimes i get very scared over something	According to my analysis, you have been potentially suffered from Phobia.	Phobia
3	so these couple days, i have a very bad headache, i had it since i was bitten by my friend dog	According to my analysis, you have been potentially suffered from Rabies.	Rabies
4	im not sick but i see someone acting irrationally, i see that he is out of control, maybe he is crazy	Sorry, I have tried my best and didn't know the disease you have. I recommend you to go to the hospital.	Crazy
5	my nose hurt, i feel cold for no reason	According to my analysis, you have been potentially suffered from Flu.	Flu
6	my throat hurts, im pretty sure i have a sore throat, sore throat	The suggested treatment for sorethroat is You should drink lots of water, consume soft food, don't talk too much, gargle with salt water, and if the sore throat is caused by bacteria, you	Sore Throat

Licensed use limited to: Alvas Institute of Engineering and Technology Department of Library and Information Centre. Downloaded on March 30,2026 at 04:31:08 UTC from IEEE Xplore. Restric

		should go to the doctor to get antibiotic medical treatment.	
7	help, my friend is dropped down to the floor and he seizure, epilepsy	The suggested treatment for epilepsy is Consume antiepilepsi medicine such as carbamazepine, clobazam, levetiracetam, etc., to stabilize your electrical neuron activity in your brain.	Epilepsy
8	my mouth is dry, and also i always feel thirsty	According to my analysis, you have been potentially suffered from Dehydration.	Dehydration

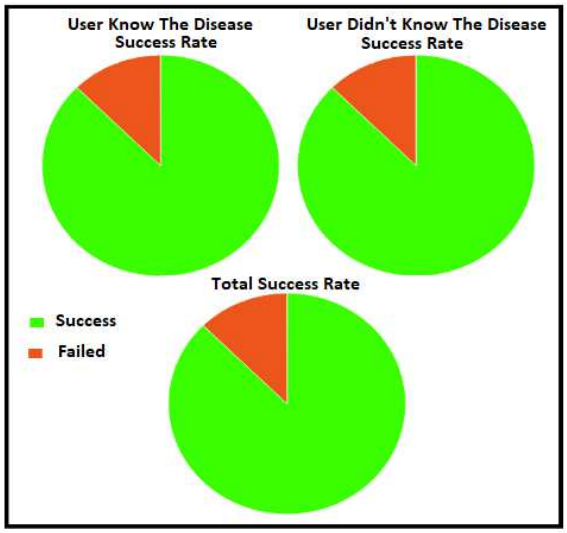


Fig. 3. Visual Representation Of Results

As shown in Figure 3. We can see that our chatbot have a high success rate for analysing the diseases from both case where the user already knows their disease and the user that didn't know what their disease is.

The success percentage from the case where user already know their disease is 87.5%, and from the case where user didn't know their disease is also 87.5%. Which made the total success rate is 87.5%.

V. Conclusions

Medical chatbots can be very useful and important for the users if they can't afford nor have the time to go to the hospital to do some consultation with the doctors about their illness. Our chatbot uses the concept of cosine similarity and ID3 decision tree to provide the diagnosis result of the disease suffered by the user.

Program accuracy is very important, but it is near impossible to reach 100% accuracy. Based on the results using the test data sets, the accuracy of our program if the user already knows their disease is 87.5%, and for the case where the user didn't know about their disease, the accuracy is 87.5%. We find that the accuracy of the program highly depends on 2 variables, which are the result of the stopwords removal process and the amount of trained data in the database.

For future work, we target to improve the chatbot accuracy and later on more features such as buying meds with the application, recommended clinic or hospital near the user, and to cooperate with doctors among the world so that when the chatbot couldn't find a proper treatment, it could automatically suggest a recommended doctor to make the user able to do further consultation.

References

[1] D. Shnavi, et al., "A Self-Diagnosis Medical Chatbot Using Artificial Intelligence", Journal of Web Development and Web Designing, vol.3, pp. 1, MAT Journals, 2018.

[2] S. Ghare, et al., "Self-Diagnosis Medical Chat-Bot Using Artificial Intelligence", pp. 1, February 2020.

[3] D. Zumstein and S. Hundertmark, "Chatbots – An Interactive Technology for Personalized Communication, Transactions and Services," IADIS International Journal on WWW/Internet, vol. 15, pp. 98, November 2017.

[4] G. Prasad K. C. et al., "A Personalized Medical Assistant Chatbot: MediBot," International Journal of Science Technology & Engineering, vol. 5, pp. 43, January 2019.

[5] D. Madhu, et al., "A novel approach for medical assistance using trained chatbot," in International Conference on Inventive Communication and Computational Technologies, pp. 1, March 2017.

[6] N. Haristiani, "Artificial Intelligence (AI) Chatbot as Language Learning Medium: An inquiry," in International Conference on Education, Science and Technology, pp. 1-5, March 2019.

[7] T. Nadarzynski, et al., "Acceptability of artificial intelligence (AI)-led chatbot services in healthcare: A mixed-methods study," Digital Health, vol. 5, pp. 1, January 2019.

[8] M. Adam, M. Wessel, and A. Benlian, "AI-based chatbots in customer service and their effects on user compliance," in The International Journal on Networked Business, pp. 1, February 2020.

[9] R. B. Mathew, et al., "Chatbot for Disease Prediction and Treatment Recommendation using Machine Learning," in International Conference on Trends in Electronics and Informatics, pp. 853, October 2019.

[10] L. Athota, et al., "Chatbot for Healthcare System Using Artificial Intelligence," in International Conference on Reliability, Infocom Technologies and Optimization, pp. 620, June 2020.

[11] C. S. Kulkrani, et al., "BANK CHAT BOT – An Intelligent Assistant System Using NLP and Machine Learning," in International Research Journal of Engineering and Technology, vol. 4, pp. 2375, May 2017.

[12] B. Su, et al., "Health Care Spoken Dialogue System for Diagnostic Reasoning and Medical Product Recommendation," in International Conference on Orange Technologies, pp. 4, October 2018.

[13] M. Virkar, V. Honmane, and S. U. Rao, "Humanizing the Chatbot with Semantics based Natural Language Generation," in International Conference on Intelligent Computing and Control Systems, pp. 893, May 2019.

[14] F. M. J. M. Shamrat, et al., "Performance evaluation among ID3, C4.5, and CART Decision Tree Algorithms," in International Conference on Pervasive Computing and Social Networking, pp. 4, March 2021.

Licensed use limited to: Alvas Institute of Engineering and Technology Department of Library and Information Centre. Downloaded on March 30,2026 at 03:31:08 UTC from IEEE Xplore. Restric

[15] S. Mujeeb, M. H. Javed, and T. Arshad, "Aquabot: A Diagnostic Chatbot for Achluophobia and Autism," in International Journal of Advanced Computer Science and Applications, pp. 209-216, January 2017.

[16] A. R. Panhalkar and D. D. Doye, "Optimization of decision trees using modified African buffalo algorithm," in Journal of King Saud University –Computer and Information Sciences, pp. 1, February 2021.

Licensed use limited to: Alvas Institute of Engineering and Technology Department of Library and Information Centre. Downloaded on March 30,2025 at 13:31:08 UTC from IEEE Xplore. Restric